

Alvaro Pastor

Professor, Master's Degree in Videogame Design and Development

Department of Computer Science, Multimedia, and Telecommunication

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1. EDUCATION

Ph.D., Network and Information Technologies

Open University of Catalonia, Spain

2026 (expected)

Dissertation: "Virtual and Augmented Reality Investigations of Episodic Memory: The Role of Locomotion, Spatial Features and Emotion"

M.Sc., Cognitive Systems and Interactive Media

Pompeu Fabra University, Spain

2015

Architect

Applied Sciences University, Peru

2013

2. PROFESSIONAL APPOINTMENTS

Professor, Master's Degree in Videogame Design and Development

Open University of Catalonia, Spain.

February 2020 – Present.

Specialist Lead, Virtual Reality Design and Development

EventLAB, Department of Clinical Psychology and Psychobiology, Universitat de Barcelona, Spain.

February 2023 – December 2024.

Professor, Degree in Videogame Design and Development

Centre for Image and Multimedia Technology, Universitat Politècnica de Catalunya, Spain.

February 2022 – July 2024.

3. RESEARCH FUNDING & LEADERSHIP

- **NeuroScent VR:** Research Accelerator Grant RA405, Universitat Oberta de Catalunya. (2024 – 2025). *Role: Lead developer.*
- **EnvironMENTAL:** Horizon Europe Grant 101057429 (2023–2024). *Role: Lead Virtual Reality design & development.*

4. SCIENTIFIC CONTRIBUTIONS

Selected Peer-Reviewed Publications:

- **Pastor, A.; Bourdin-Kreitz, P. (2026). Designing memorable XR under stress: An empirically grounded criterion for long-term memory binding.** In CHI 2026 Workshop on XR for Challenging Environments (XR4CE): Enabling Human Performance and Agency under Stress. ACM.
- **Pastor, A.; Bourdin-Kreitz, P. (2025, *in review*). Synthetic generation of photorealistic 3D characters for face-based episodic memory measures.** *Transactions on Applied Perception*. https://doi.org/10.31219/osf.io/4jf2g_v1
- **Pastor, A.; Bourdin-Kreitz, P. (2025, *in press*). Comparing Episodic Memory Binding Outcomes After Emotion Induction in Virtual Reality.** *Virtual Reality*.
- **Pastor, A.; Bourdin-Kreitz, P. (2024). Comparing episodic memory outcomes from walking augmented reality and stationary virtual reality encoding experiences.** *Scientific Reports*, 14, 1-23. <https://doi.org/10.1038/s41598-024-57668-w>

Selected Research Projects:

- **NeuroScent VR Project (Research Accelerator Grant RA405, Open University of Catalonia, 2024–2025).** *Role: Lead developer.*

- **EnvironMENTAL Project** (Horizon Europe Grant 101057429, Universitat de Barcelona, 2023–2024). *Role: Lead technical specialist for Virtual Reality experiments.*
- **The Memory Palace Project** (Innovative Information and Communication Technologies Grant, Barcelona Culture Institute, 2021–2022). *Role: Principal Investigator.*

5. TECHNICAL & METHODOLOGICAL EXPERTISE

- **Immersive & interactive Systems:** AR/VR design and development. Interactive systems engineering. Full-stack interactive web technologies. User experience optimisation for immersive environments.
- **Cognitive science methods:** Creation of ecologically valid measurement frameworks for immersive environments. Development and validation of novel cognitive assessments. Integration of standard psychometrics into experimental designs.
- **Machine learning & computer vision:** Neural network-based 2D/3D synthesis. Computer vision applications. Algorithmic generation of auditory, semantic and visual stimuli.
- **Hardware integration:** Psychophysiological sensor integration in immersive environments. Electromechanical systems engineering for experimental platforms. Multimodal data synchronisation.
- **Data science & computational methods:** Statistical analysis and computational modeling. Behavioural and physiological data processing. Scientific data visualisation.